ME 3624
Design Project Report – Fall 2022

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"I have neither given nor received unauthorized assistance on this project."

Initials/Signatures: _____________IA_________________

Contents

Project Summary

Introduction

- The objective of the FEA Project is to perform static analysis of a mechanical beam structure by applying statics, deforms, and mechanical design concepts and to compare the derived analytical solutions with a FEA study using CAD in Solidworks.
- The material properties used in this assignment were the Modulus of Elasticity (E) , the density (ρ) and the Yield Strength (Sy).
- To calculate the Factor of Safety, the Distortion Energy (DE) approximation formula was used. The small angle assumption was also implemented to compute the rotation angle θ.

Engineering design calculations

1. Given information was laid out including material properties and weight of both beams
2. The cross sectional area (A) along with the second moment of area with respect to the Z-axis were calculated using the following formulas respectfully: $A = \pi R^2$ and $I = \frac{\pi}{64}(Dout^2 - Din^2)$
3. All forces and moments acting on the fixed end were computed: $M = Force \cdot Moment\ Arm$
4. A cross section at the critical point (fixed end) was drawn and the bending, torsional, transverse, and axial stress were calculated. Since there were no torques or transverse forces the transverse and torsional shear stress were zero, so the following formulas for the axial and bending stresses were used: $\sigma\ axial = \frac{P}{A}$ and $\sigma\ bending\ = \frac{M \cdot y}{I}$ and the weights of both beams were calculated using the following formulas: $weight\ bc\ = \rho \times A \times L$; $weight\ ab = \rho \times A \times H$
5. The Factor of Safety was calculated using the DE approximation formula: $FOS = \frac{Sy}{\sigma\ max}$
6. The maximum deflection which occurs at point c was calculated using the following formulas: $M = Fl Deflection\ from\ \theta = \frac{M \cdot H \cdot L}{E \cdot I}$; $Deflection\ from\ Force\ F = \frac{F \times L^3}{3 \times E \times I}$; $Deflection\ of\ bc = \frac{wl^4}{8EI}$ $Deflection\ of\ ab = \frac{MH^2}{2EI}$; $y\ max\ = \Sigma\ all\ deflections$

Finite element model

1. A new part was created
2. A sketch of appropriate shape was drawn and dimensioned, and then extruded along the y-axis with length H to create beam AB
3. An extruded cut of appropriate dimensions was applied to the whole length of beam ab
4. Another sketch of appropriate shape was drawn using the right plane (x-axis out of plane) on the center of the top edge of beam AB then unmerged from Beam AB and extruded along that direction with length L to create beam BC
5. Another extruded cut of appropriate dimensions was applied along the length of beam BC
6. A new static analysis study was created and named through simulation->study->static
7. The beams were set to AISI 1045 CD Steel through "Apply material", and then selected and set to "treat as beams"
8. A fully fixed fixture was created using "fixed geometry" at the beginning of Beam AB
9. External loads, gravity and 100 lbf were applied using "External loads" in the appropriate directions.
10. A mesh was created and then the test was run.
11. The units and stress/deflection parameters were adjusted using the bottom most tabs that after the test was run

Baseline analysis summary

- The baseline design is clearly very safe and has a small reflection as per the requirements imposed by the manufacturer. And that makes sense since the weight is relatively larger than other options by a good margin and the second moment of area is large enough to support the structure from bending drastically
- The calculations and the FEA results prove quite similar as demonstrated in Table 1. The factor of safety calculations were almost exact as the bending stresses and the used value for Sy were pretty close. The maximum deflection calculations however have a slightly larger discrepancy, and that may be due to round off errors, and the small angle assumption.
- As discussed earlier it can be clearly seen that the baseline design is very safe and robust because of the higher second moment of area value and the relatively higher total weight compared to other design options, From that we can conclude the most important aspects of design are weight, the second moment of area, and the cross sectional area. Changing the weight can be challenging but changing the second moment of area is a simple and powerful technique that can affect the stability and robustness of the structure.

Re-design analysis summary

- As seen earlier in the formula for the weights of both beams, the cross-sectional area is directly proportional to the weight, and thus the cross sectional area should be taken into account in the redesign process since minimizing it minimizes the weight. Another design parameter is the second moment or area. Through appropriate orientation it is possible to maximize the second moment of area without changing any material properties and dimensions. The direct effect of that change is seen in the bending moment formula where the second moment of inertia (I) is inversely proportional to the bending stresses, so by decreasing it, the FOS increases since Sy is constant. The deflection also decreases with increased values of (I) as most deflection formulas contain (I) in the denominator.
- Through the calculations it is evident that some design options were simply unacceptable since they all deflected outside the maximum possible deflection limit set by the manufacturer so those were eliminated from the analysis. From this process of elimination and comparing, the design with the lowest weight given the FOS and maximum deflection values were satisfied is Design option E.

Discussion

- Through relating mechanical design and deforming phenomena and parameters to mathematical relations, and through trial and error of different redesign options, we conclude that the main design parameters that must be carefully tuned are the cross-sectional area and the area moment of inertia. Those two should be optimized to decrease the maximum combined stress applied on the structure and decrease the weight/ material cost as much as possible without sacrificing safety and overall stability.
- Engineering analysis on its own was sufficient as the problem is quite straightforward and statically determinate. Therefore, proper knowledge of the strength of materials and material properties can directly lead to a solution.
- Material science research is a powerful method for improving structure design and safety. Coating surfaces with appropriate chemicals and composites can save material cost, improve strength, and reduce costs. Another aspect to look at are different geometries and simulations. For example, the use of I -shaped beams is a powerful technique because they are designed in a way to minimize stress and material cost simply by a smart way of load distribution furthest possible from the neutral axis of bending. Trusses are another example of excellent use of material designs and solid mechanics.

Tables

Table 1: Redesign analysis summary

		Baseline Round tube, OD 6”, 0.125		Redesign *Rectangular tube 4”x2”,0.125*	
	Units	Design Eqns.	FEA	Design Eqns.	FEA
Max. deflection	[in]	1.768	1.734	5.844	5.375
Factor of Safety	[-]	14.909	14.889	7.553	7.398
Weight	[lb]	196.566	N/A	122.475	N/A

Figures

FE simulations Setup

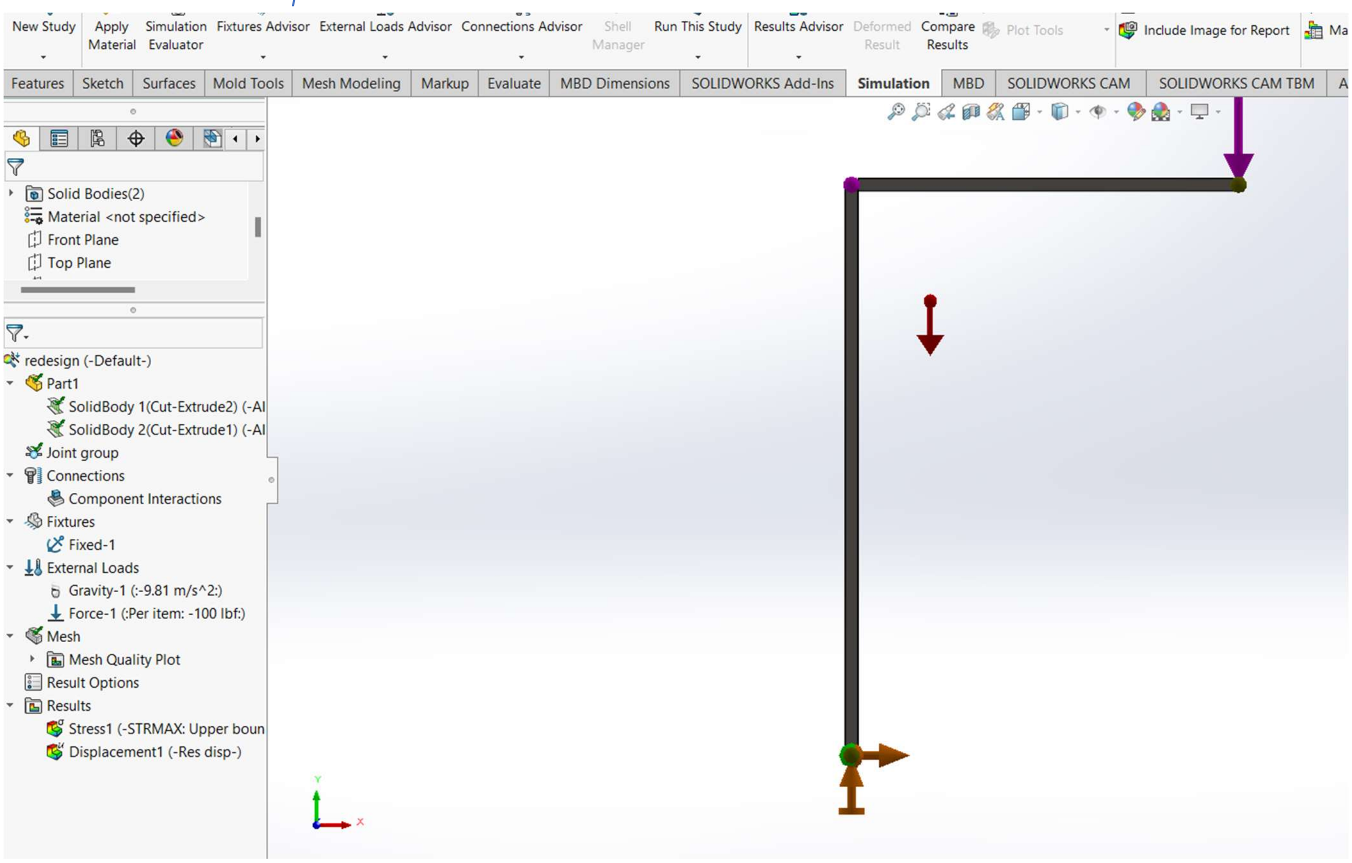


Figure 1: Boundary conditions and External Loads used for the analysis.

Baseline Model Analysis

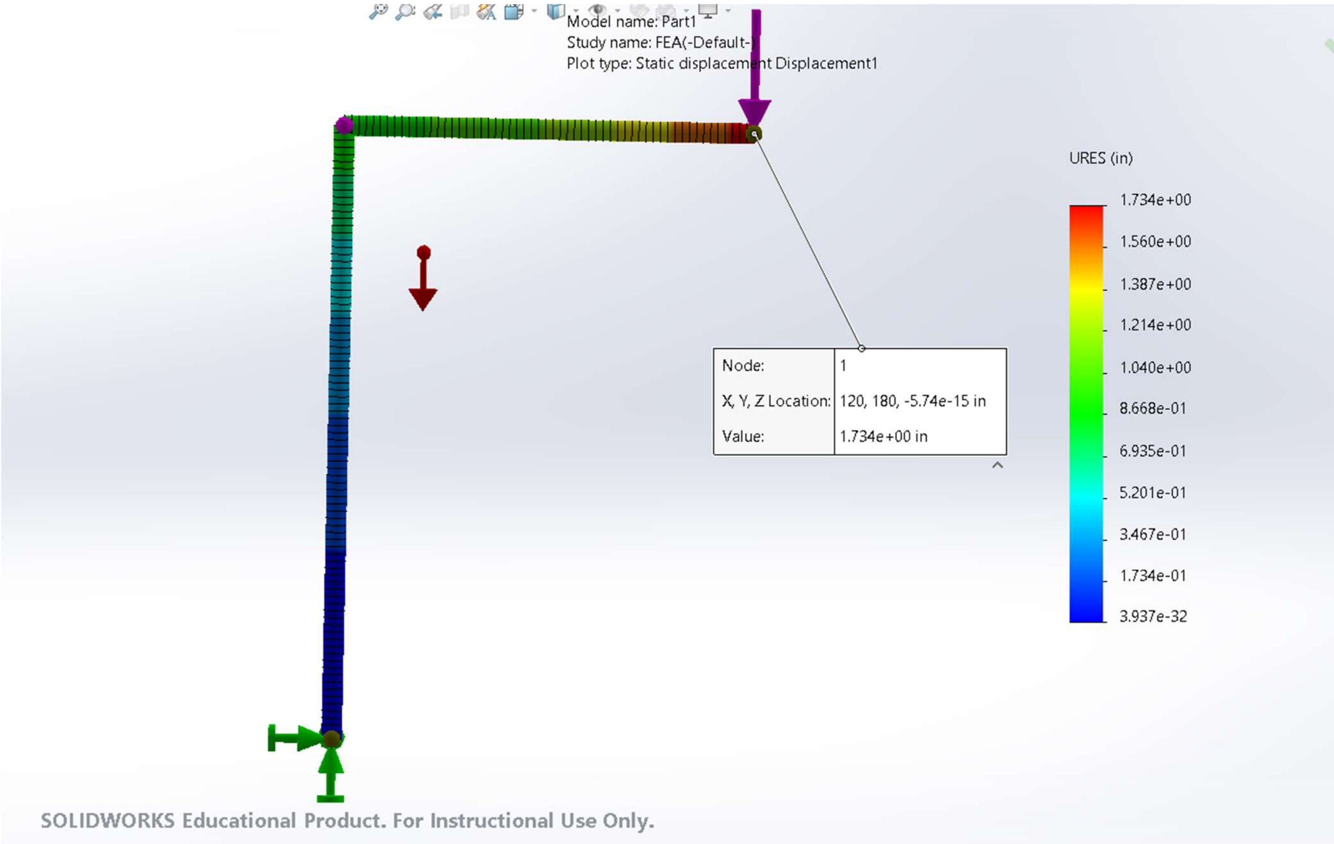


Figure 2: Maximum deflection measured on the structure.

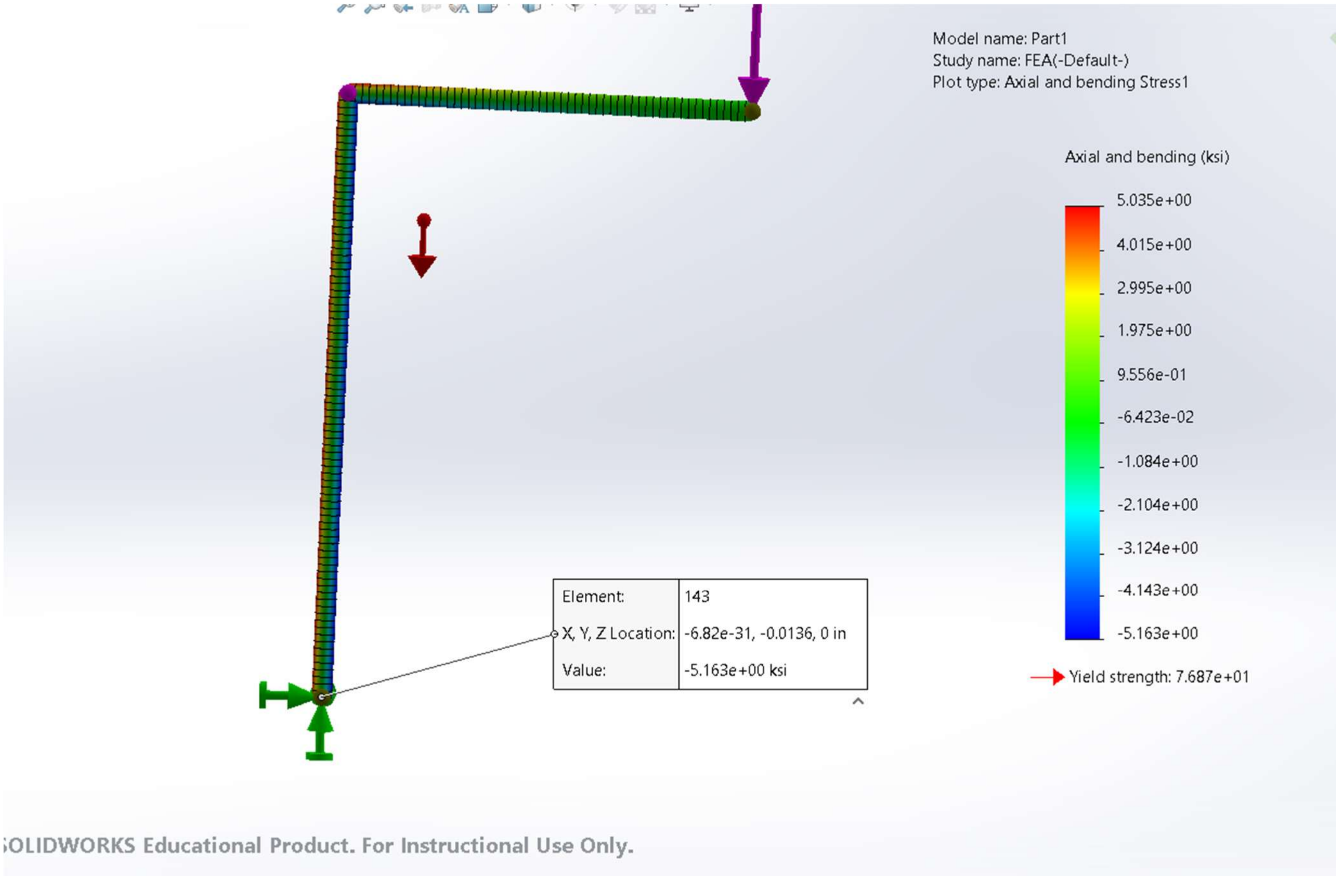


Figure 3: Stress plot showing the stresses at the critical point used to calculate the factor of safety.

Re-designed Model Analysis

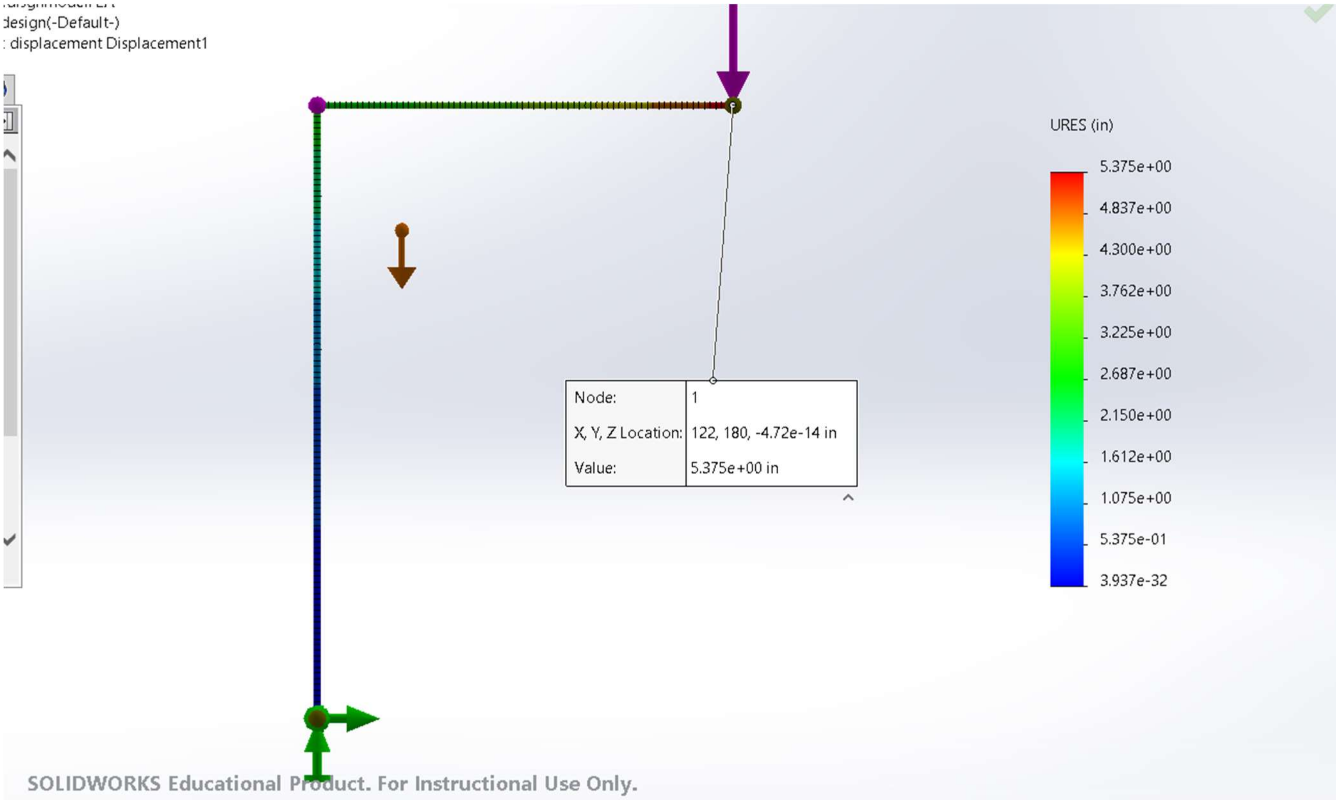


Figure 4: Maximum deflection measured on the structure.

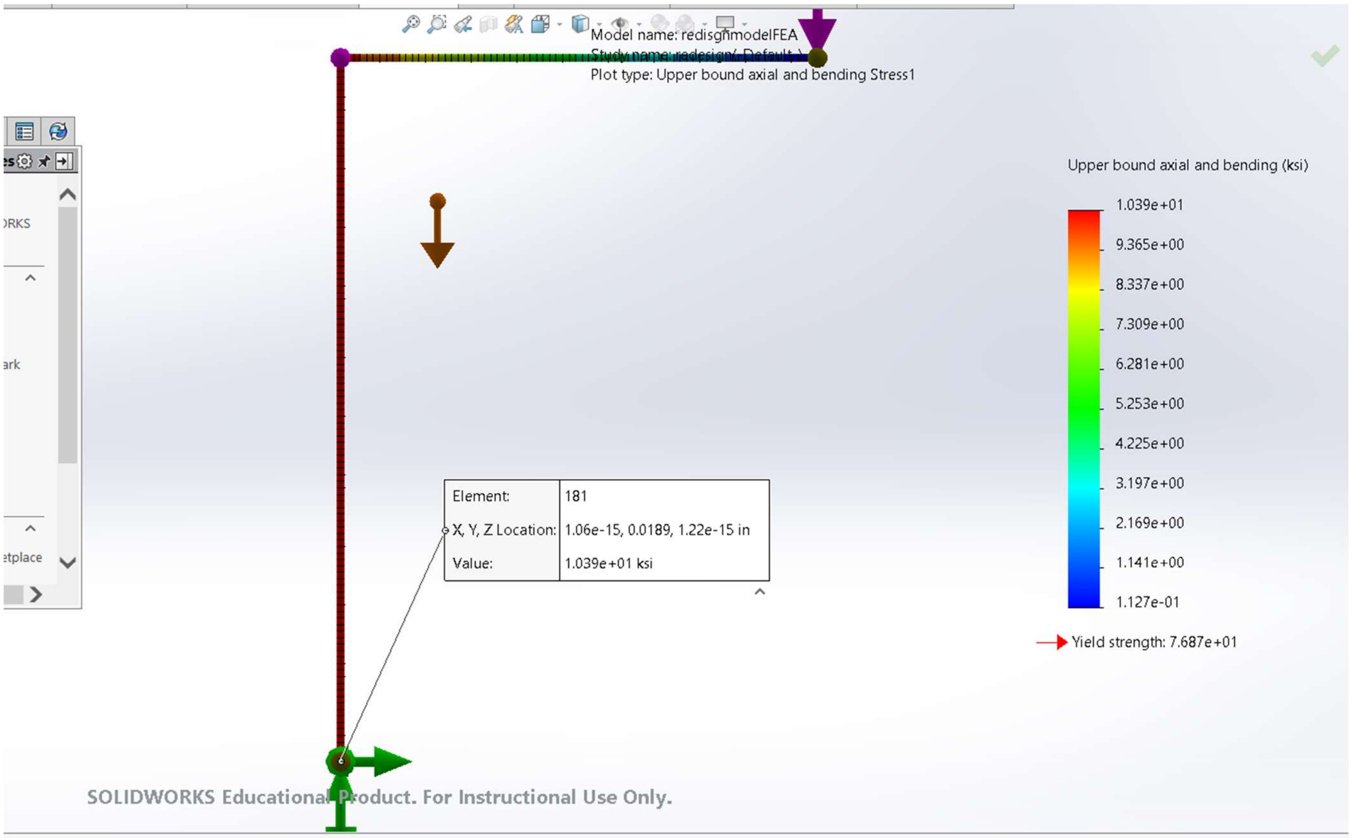
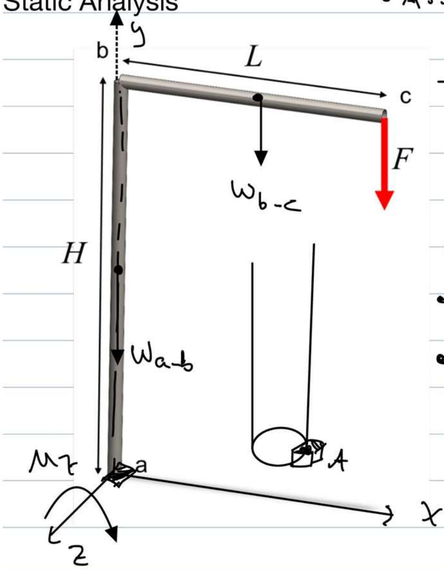


Figure 5: Stress plot showing the stresses at the critical point used to calculate the factor of safety.

Appendix

Baseline model analysis

Static Analysis



• Assume Bar b–c ⊥ Bar a–b

1. Given:

- $L = 10\,ft$
- $H = 15\,ft$
- $D_{out} = 6\,in$
- $t = 0.125\,in$
- Material AISI CD steel.
- $D_{in} = 6 - 2(0.125) = 5.75\,in$
- $F = 100\,lbf$
- $S_y = 77\,ksi$

2. Find A, I, J

$$A = \frac{\pi}{4}(D_{out}^2 - D_{in}^2) = \frac{\pi}{4}(6^2 - 5.75^2) = 2.3071\,in^2$$

$$I = \frac{\pi}{64}(D_{out}^4 - D_{in}^4) = \frac{\pi}{64}(6^4 - 5.75^4)$$

$$I = 9.958\,in^4$$

$$J = \frac{\pi}{32}(D_{out}^4 - D_{in}^4) = \frac{\pi}{32}(6^4 - 5.75^4) = 19.917\,in^4$$

3. Find V, M, P

- $V_x = 0$, • $V_z = 0$
- $P_y = -W_{a-b} - W_{b-c} - F$

$W_{a-b} = \rho \cdot A_{cs} \cdot H$

$= (0.284)(2.3071)(15 \cdot 12)$

$= 117.94 \text{ lbf}$

$W_{b-c} = \rho \cdot A_{cs} \cdot L$

$= (0.284)(2.3071)(10 \cdot 12)$

$= 78.626 \text{ lbf.}$

$\therefore P_y = -117.94 - 78.626 - 100$

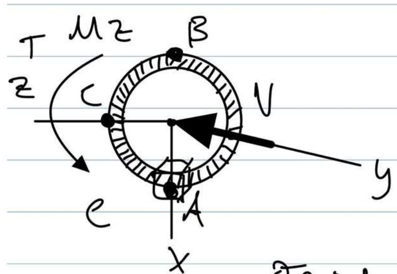
$= -296.566 \text{ lbf.}$

- $T_y = 0$
- $M_x = 0$
- $M_z = -F(L) - W_{b-c}(L/2)$

$= -100(10 \cdot 12) - 78.626(5 \cdot 12)$

$\therefore M_z = -16,717.56 \text{ lbf-in}$

4. Draw & Analyze critical x-section



$\sigma_x = 0$ $\tau_{xz} = 0$

$\sigma_y = \sigma_{axial} + \sigma_{bending}$ $\tau_{yz} = 0$

$\sigma_z = 0$ $\tau_{xy} = 0$

$\sigma_{axial} = \frac{-P}{A} = -\frac{296.566}{2.3071} = -128.545 \text{ psi}$

At pt B:

$\sigma_{bending} = 5036.12 - 128.545 = 4907.575$ psi

$$\sigma_{bending} = \frac{-M \cdot c_0}{I} = \frac{-(16,716.56)(3)}{9.958}$$

$\therefore$

$$\sigma_{bending} = -5036.12 \text{ psi}$$

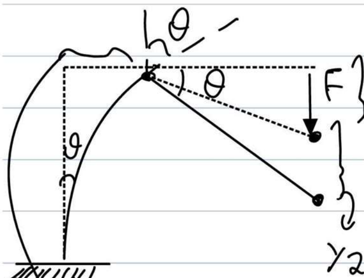
$$\sigma_y = \sigma_{max} = -5164.66 \text{ psi}$$

$$= -5.164 \text{ ksi}$$

4. Find Factor of Safety:

$$n = \frac{S_y}{|\sigma|} = \frac{77}{5.164} = 14.91$$

5. Find the deflection:



$$y_1 = \theta \cdot L \;;\; \theta = \frac{MH}{EI}$$

$$y_2 = \frac{FL^3}{3EI} + \frac{wL^4}{8EI}$$

$$x = \frac{MH^2}{2EI} \;;\; M = FL \quad \therefore \quad M = (100)(10.12) = 12{,}000 \text{ lbf-in}$$

$$\theta = \frac{(12000)(15.12)}{(30 \times 10^6)(9.958)} = 7.2304 \times 10^{-3}\,{}^\circ$$

$$\therefore y_1 = \theta \cdot L = (7.2304 \times 10^{-3})(10.12) = 0.8676 \text{ in}$$

$$y_2 = \frac{(100)(10.12)^3}{3(30\times10^6)(9.958)} + \frac{\left(\frac{78.626}{10.12}\right)(10.12)^4}{8(30\times10^6)(9.958)}$$

$$y_2 = 0.2496 \text{ in}$$

$$x = \frac{MH^2}{2EI} = \frac{(12000)(15.12)^2}{2(30\times10^6)(9.958)} = 0.6507 \text{ in}$$

$$y_{max} = y_1 + y_2 + x = 0.868 + 0.2496 + 0.6507$$
$$= 1.768 \text{ in}$$

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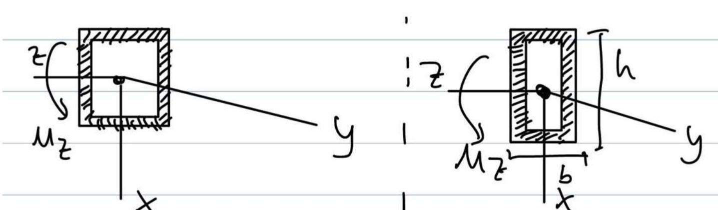
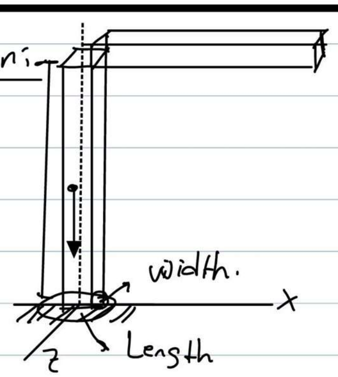
5. Find formula for I for rectangle and square cross-sections:



$$I = \frac{1}{12}\left(S_{out}^4 - S_{in}^4\right) \qquad I_z = \left(\frac{bh^3}{12}\right)_{out} - \left(\frac{bh^3}{12}\right)_{in}$$

Redesign:



Re-design process and analysis

Given	
E (psi) =	30000000
Density (lb/in^3) =	0.284
L (in) =	120
H (in) =	180
F=	100
Sy (ksi) =	77
M (lbf-in)	12000

Circular			
	Dout (in)	**t (in)**	**Din (in)**
baseline	6	0.125	5.75
a	5	0.125	4.75
b	3	0.125	2.75

Square				
	Sout		**t**	**Sin**
c	6		0.125	5.75
d	4		0.125	3.75

	lrngth out	**width out**	**t**	**length in**	**lwidth in**
e	**2**	**4**	0.125	1.75	3.75
f	**2**	**6**	0.125	1.75	5.75

Reversed Rectangle					
	lrngth out	**width out**	**t**	**length in**	**lwidth in**
e	4	2	0.125	3.75	1.75
f	6	2	0.125	5.75	1.75

Please enlarge the image if needed

Deesign	Area	I	W(b-c)	W(a-b)	Axial force (P) abs	Sigma axial abs (psi)	Mz (lbf-in) abs	Sigma bending abs (psi)	sigma max abs (ksi)
basline	2.307107105	9.958411527	78.62621014	117.9393152	296.5655253	128.5443249	16717.57261	5036.216638	5.164760963
design a	1.914408023	5.690876975	65.24302543	97.86453815	263.1075636	137.4354685	15914.58153	6991.269357	7.128704826
design b	1.12900986	1.168701613	38.47665602	57.71498404	196.1916401	173.7731857	14308.59936	18364.73811	18.53851129
design c	2.9375	16.90592448	100.11	150.165	350.275	119.2425532	18006.6	3195.317716	3.31456027
design d	1.9375	4.853841146	66.03	99.045	265.075	136.8129032	15961.8	6576.976675	6.713789578
design e	1.4375	0.991861979	48.99	73.485	222.475	154.7652174	14939.4	15061.97466	15.21673988
design f	1.9375	1.431966146	66.03	99.045	265.075	136.8129032	15961.8	11146.7719	11.28358481
Reversed Design e	1.4375	2.976236979	48.99	73.485	222.475	154.7652174	14939.4	10039.11994	10.19388516
Reversed Design F	1.9375	8.275716146	66.03	99.045	265.075	136.8129032	15961.8	5786.254525	5.923067429

Deesign	FOS	Theta	y(theta)	y(f)	y(w-b-c)	x(w a-b)	ymax	Total Weight	
basline	14.90872483	0.007230069	0.86760825	0.19280183	0.0568473	0.6507062	1.767964	196.5655253	
design a	10.80140108	0.012651829	1.51821943	0.3373821	0.0825444	1.1386646	3.07681	163.1075636	
design b	4.153515824	0.061606829	7.392819438	1.64284876	0.2370425	5.5446146	14.81733	96.19164006	
design c	23.23083418	0.004258862	0.511063445	0.11356965	0.0426355	0.3832976	1.050566	250.275	
design d	11.46893258	0.014833613	1.780033532	0.39556301	0.0979463	1.3350251	3.608568	165.075	
design e	5.060216617	0.072590745	8.710889399	1.9357532	0.3556221	6.533167	17.53543	122.475	
design f	6.824072431	0.050280518	6.033662196	1.34081382	0.3320023	4.5252466	12.23172	165.075	
Reversed Design e	7.553547918	0.024191622	2.902994641	0.64510992	0.1185148	2.177246	5.843865	122.475	Winner
Reversed Design F	13.00002084	0.008700153	1.044018409	0.23200409	0.0574471	0.7830138	2.116483	165.075	